Homework 05 - Law and courts

INFO 5001 - Fall 2025

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Setup

Load packages and data:

```r
install.packages(c("tidyverse", "rvest", "janitor"))
```

```
The following package(s) will be installed:
- janitor   [2.2.1]
- rvest     [1.0.5]
- tidyverse [2.0.0]
These packages will be installed into "~/.local/hw-05-jl4495-1/renv/library/
R-4.3/x86_64-pc-linux-gnu".

# Installing packages --------------------------------------------------------
- Installing rvest ...                          OK [linked from cache]
- Installing tidyverse ...                      OK [linked from cache]
- Installing janitor ...                        OK [linked from cache]
Successfully installed 3 packages in 25 milliseconds.
```

```r
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0
──
✔ dplyr     1.1.4     ✔ readr     2.1.5
✔ forcats   1.0.1     ✔ stringr   1.5.2
✔ ggplot2   4.0.0     ✔ tibble    3.3.0
✔ lubridate 1.9.4     ✔ tidyr     1.3.1
✔ purrr     1.1.0
── Conflicts ──────────────────────────────────────────── tidyverse_conflicts()
──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
library(scales)
```

```

Attaching package: 'scales'

The following object is masked from 'package:purrr':

    discard

The following object is masked from 'package:readr':

    col_factor
```

```
library(rvest)
```

```

Attaching package: 'rvest'

The following object is masked from 'package:readr':

    guess_encoding
```

Exercise 1

```
library(rvest)
library(tidyverse)
library(janitor)
```

```

Attaching package: 'janitor'
```

```
The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```
url <- "https://www.prisonpolicy.org/phones/appendices2022_1.html"
tables <- read_html(url) |> html_table(fill = TRUE)
phone_raw <- tables[[1]]
```

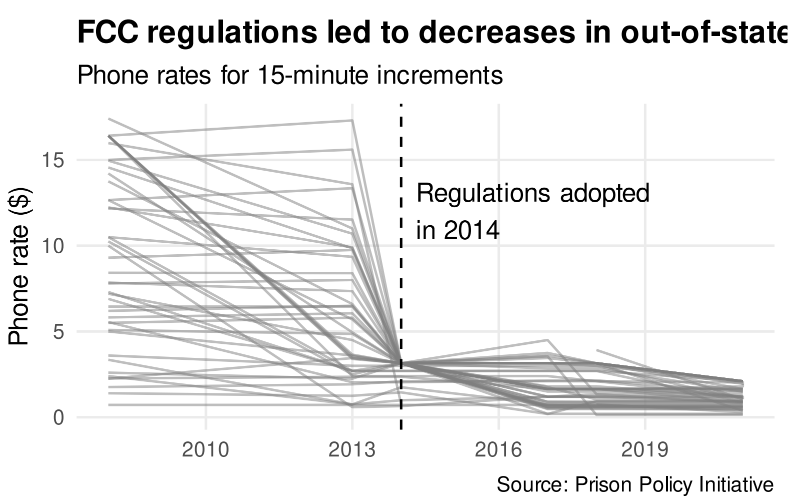
```r
header1 <- phone_raw[1, ]
header2 <- phone_raw[2, ]
new_header <- ifelse(header2 == "", header1, paste(header1, header2, sep =
"_"))
colnames(phone_raw) <- new_header

phone_raw <- phone_raw[-c(1, 2), ]
phone_raw <- janitor::clean_names(phone_raw)
out_state_raw <- phone_raw[, c(1, 9:ncol(phone_raw))]

out_state_tidy <- out_state_raw |>
  rename(states = 1) |>
  pivot_longer(
    cols = -states,
    names_to = "year",
    values_to = "rate"
  ) |>
  mutate(
    type = "out_state",
    year = readr::parse_number(year),
    rate = readr::parse_number(rate)
  )
```

```
Warning: There were 2 warnings in `mutate()`.
The first warning was:
ℹ In argument: `year = readr::parse_number(year)`.
Caused by warning:
! 49 parsing failures.
row col expected actual
  2  -- a number     na
 10  -- a number     na
 18  -- a number     na
 26  -- a number     na
 34  -- a number     na
... ... ........ ......
See problems(...) for more details.
ℹ Run `dplyr::last_dplyr_warnings()` to see the 1 remaining warning.
```

```r
ggplot(out_state_tidy, aes(x = year, y = rate, group = states)) +
  geom_line(alpha = 0.5, color = "gray50") +
  geom_vline(xintercept = 2014, linetype = "dashed") +
  annotate(
    "text",
    x = 2014.3,
    y = 12,
```

```
    label = "Regulations adopted\nin 2014",
    hjust = 0
  ) +
  labs(
    title = "FCC regulations led to decreases in out-of-state prison phone
rates",
    subtitle = "Phone rates for 15-minute increments",
    x = NULL,
    y = "Phone rate ($)",
    caption = "Source: Prison Policy Initiative"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold"),
    panel.grid.minor = element_blank()
  )
```

```
Warning: Removed 51 rows containing missing values or values outside the scale
range
(`geom_line()`).
```



Exercise 2

```
scdb_case <- read_csv("data/scdb-case.csv")
```

```
Rows: 29227 Columns: 52
── Column specification
────────────────────────────────────────────────────────
Delimiter: ","
```

```
chr (14): caseId, docketId, caseIssuesId, dateDecision, usCite, sctCite,
led...
dbl (38): decisionType, term, naturalCourt, petitioner, petitionerState,
res...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

```
scdb_vote <- read_csv("data/scdb-vote.csv")
```

```
Rows: 255857 Columns: 13
── Column specification
────────────────────────────────────────────────────────
Delimiter: ","
chr (5): caseId, docketId, caseIssuesId, voteId, justiceName
dbl (8): term, justice, vote, opinion, direction, majority,
firstAgreement, ...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

```
scdb_joined <- scdb_case |>
  left_join(scdb_vote, by = "caseId")
glimpse(scdb_joined)
```

```
Rows: 255,857
Columns: 64
$ caseId                  <chr> "1791-001", "1791-001", "1791-001",
"1791-001…
$ docketId.x              <chr> "1791-001-01", "1791-001-01", "1791-001-01",
…
$ caseIssuesId.x          <chr> "1791-001-01-01", "1791-001-01-01",
"1791-001…
$ dateDecision            <chr> "8/3/1791", "8/3/1791", "8/3/1791",
"8/3/1791…
$ decisionType            <dbl> 6, 6, 6, 6, 6, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
…
$ usCite                  <chr> "2 U.S. 401", "2 U.S. 401", "2 U.S. 401", "2
…
$ sctCite                 <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
$ ledCite                 <chr> "1 L. Ed. 433", "1 L. Ed. 433", "1 L. Ed.
```

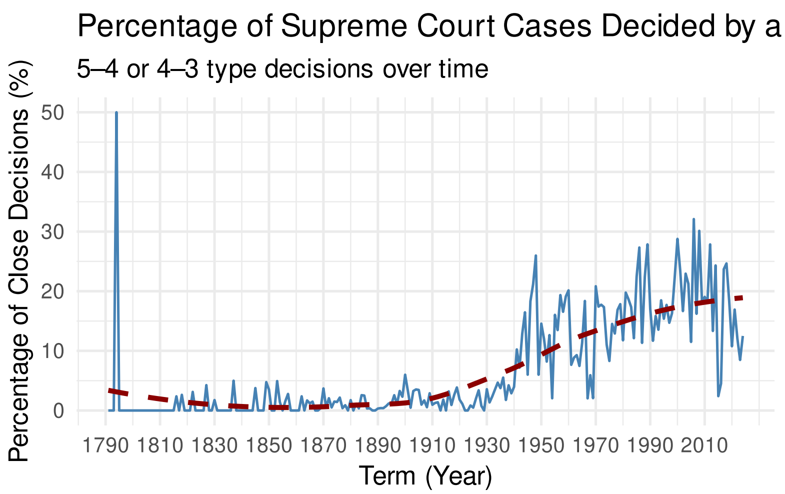
```
433…
$ lexisCite                <chr> "1791 U.S. LEXIS 189", "1791 U.S. LEXIS
189",…
$ term.x                   <dbl> 1791, 1791, 1791, 1791, 1791, 1791, 1791,
179…
$ naturalCourt             <dbl> 102, 102, 102, 102, 102, 102, 102, 102, 102,
…
$ chief                    <chr> "Jay", "Jay", "Jay", "Jay", "Jay", "Jay",
"Ja…
$ docket                   <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
$ caseName                 <chr> "WEST, PLS. IN ERR. VERSUS BARNES. et al.",
"…
$ dateArgument             <chr> "8/2/1791", "8/2/1791", "8/2/1791",
"8/2/1791…
$ dateRearg                <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
$ petitioner               <dbl> 501, 501, 501, 501, 501, 501, 501, 501, 501,
…
$ petitionerState          <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
$ respondent               <dbl> 111, 111, 111, 111, 111, 28, 28, 28, 28, 28,
…
$ respondentState          <dbl> NA, NA, NA, NA, NA, 25, 25, 25, 25, 25, 37,
3…
$ jurisdiction             <dbl> 13, 13, 13, 13, 13, 15, 15, 15, 15, 15, 15,
1…
$ adminAction              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
$ adminActionState         <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
$ threeJudgeFdc            <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
…
$ caseOrigin               <dbl> 431, 431, 431, 431, 431, NA, NA, NA, NA, NA,
…
$ caseOriginState          <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
$ caseSource               <dbl> 431, 431, 431, 431, 431, NA, NA, NA, NA, NA,
…
$ caseSourceState          <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
$ lcDisagreement           <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
…
$ certReason               <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
…
$ lcDisposition            <dbl> NA, NA, NA, NA, NA, 9, 9, 9, 9, 9, NA, NA,
NA…
$ lcDispositionDirection   <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
```

```
…
$ declarationUncon         <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
…
$ caseDisposition          <dbl> 9, 9, 9, 9, 9, 1, 1, 1, 1, 1, 9, 9, 9, 9, 9,
…
$ caseDispositionUnusual   <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
…
$ partyWinning             <dbl> 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0,
…
$ precedentAlteration      <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
…
$ voteUnclear              <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
…
$ issue                    <dbl> 90320, 90320, 90320, 90320, 90320, 90320,
903…
$ issueArea                <dbl> 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9,
…
$ decisionDirection        <dbl> 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 1, 1, 1, 1, 1,
…
$ decisionDirectionDissent <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
…
$ authorityDecision1       <dbl> 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3,
…
$ authorityDecision2       <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
$ lawType                  <dbl> 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 6, 6, 6, 6, 6,
…
$ lawSupp                  <dbl> 403, 403, 403, 403, 403, 403, 403, 403, 403,
…
$ lawMinor                 <chr> "NULL", "NULL", "NULL", "NULL", "NULL",
"NULL…
$ majOpinWriter            <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
$ majOpinAssigner          <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
…
$ splitVote                <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
…
$ majVotes                 <dbl> 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 6, 6, 6, 6, 6,
…
$ minVotes                 <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
…
$ docketId.y               <chr> "1791-001-01", "1791-001-01", "1791-001-01",
…
$ caseIssuesId.y           <chr> "1791-001-01-01", "1791-001-01-01",
"1791-001…
$ voteId                   <chr> "1791-001-01-01-01-01",
"1791-001-01-01-01-02…
$ term.y                   <dbl> 1791, 1791, 1791, 1791, 1791, 1791, 1791,
```

```
179…
$ justice                    <dbl> 1, 3, 4, 5, 6, 1, 3, 4, 5, 6, 1, 3, 4, 5, 6,
…
$ justiceName                <chr> "JJay", "WCushing", "JWilson", "JBlair",
"JIr…
$ vote                       <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
…
$ opinion                    <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
…
$ direction                  <dbl> 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 1, 1, 1, 1, 1,
…
$ majority                   <dbl> 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
…
$ firstAgreement             <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
$ secondAgreement            <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N…
```

```
one_vote_margin <- scdb_joined |>
  filter(!is.na(majVotes), !is.na(minVotes)) |>
  mutate(one_vote_margin = abs(majVotes - minVotes) == 1)

one_vote_summary <- one_vote_margin |>
  group_by(term.y) |>
  summarise(
    total_cases = n(),
    one_vote_margin_cases = sum(one_vote_margin, na.rm = TRUE),
    pct_one_vote = (one_vote_margin_cases / total_cases) * 100
  )

  ggplot(one_vote_summary, aes(x = term.y, y = pct_one_vote)) +
  geom_line(color = "steelblue") +
  geom_smooth(method = "loess", se = FALSE, color = "darkred", linetype =
"dashed") +
  labs(
    title = "Percentage of Supreme Court Cases Decided by a One-Vote Margin",
    subtitle = "5–4 or 4–3 type decisions over time",
    x = "Term (Year)",
    y = "Percentage of Close Decisions (%)"
  ) +
  scale_x_continuous(breaks = seq(1790, 2020, by = 20)) + # makes term axis
continuous
  theme_minimal(base_size = 13)
```

```
`geom_smooth()` using formula = 'y ~ x'
```




```
# According to the graph, the percentage of on-vote margin starts to gradually increase from the starting from 1895.
```

Exercise 3

```
library(tidyverse)

scdb_case <- read_csv("data/scdb-case.csv")
```

```
Rows: 29227 Columns: 52
── Column specification
────────────────────────────────────────────────────────
Delimiter: ","
chr (14): caseId, docketId, caseIssuesId, dateDecision, usCite, sctCite,
led...
dbl (38): decisionType, term, naturalCourt, petitioner, petitionerState,
res...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

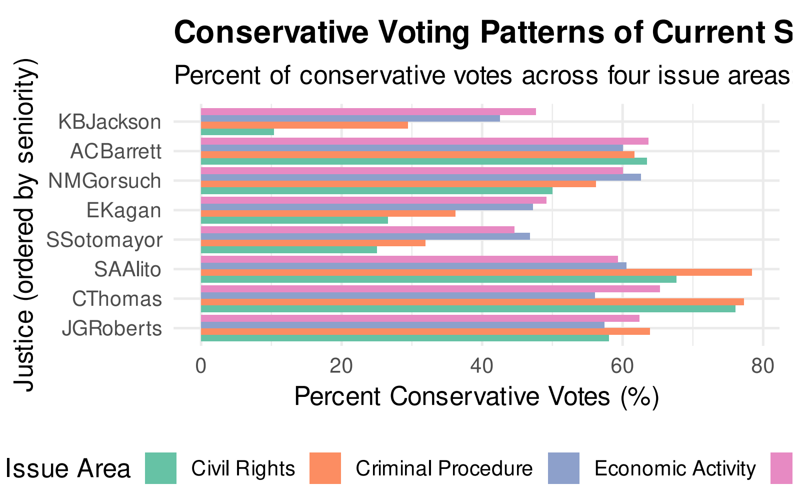
```
scdb_vote <- read_csv("data/scdb-vote.csv")
```

```
Rows: 255857 Columns: 13
── Column specification
────────────────────────────────────────────────────────
Delimiter: ","
```

```
chr (5): caseId, docketId, caseIssuesId, voteId, justiceName
dbl (8): term, justice, vote, opinion, direction, majority,
firstAgreement, ...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

```
scdb_joined <- scdb_vote |>
  left_join(scdb_case |> select(caseId, issueArea, term), by = "caseId")

scdb_joined <- scdb_joined |>
  mutate(
    issueAreaLabel = recode(as.character(issueArea),
      "1" = "Criminal Procedure",
      "2" = "Civil Rights",
      "8" = "Economic Activity",
      "9" = "Judicial Power",
      .default = NA_character_
    )
  )

scdb_filtered <- scdb_joined |>
  filter(!is.na(issueAreaLabel), !is.na(direction))

scdb_filtered <- scdb_filtered |>
  mutate(conservative_vote = if_else(direction == 1, 1, 0))

current_justices <- tibble(
  justiceName = c(
    "JGRoberts",
    "CThomas",
    "SAAlito",
    "SSotomayor",
    "EKagan",
    "NMGorsuch",
    "BKavanaugh",
    "ACBarrett",
    "KBJackson"
  ),
  seniority_rank = 1:9
)

conservative_summary <- scdb_filtered |>
  filter(justiceName %in% current_justices$justiceName) |>
  group_by(justiceName, issueAreaLabel) |>
  summarise(
```

```
    total_votes = n(),
    conservative_votes = sum(conservative_vote, na.rm = TRUE),
    pct_conservative = (conservative_votes / total_votes) * 100,
    .groups = "drop"
  ) |>
  left_join(current_justices, by = "justiceName")

ggplot(conservative_summary, aes(
  x = reorder(justiceName, seniority_rank),
  y = pct_conservative,
  fill = issueAreaLabel
)) +
  geom_col(position = "dodge") +
  coord_flip() +
  scale_fill_brewer(palette = "Set2") +
  labs(
    title = "Conservative Voting Patterns of Current Supreme Court Justices",
    subtitle = "Percent of conservative votes across four issue areas:
Criminal Procedure, Civil Rights, Economic Activity, and Judicial Power",
    x = "Justice (ordered by seniority)",
    y = "Percent Conservative Votes (%)",
    fill = "Issue Area"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold"),
    legend.position = "bottom"
  )
```



Exercise 4

```r
library(tidyverse)
library(lubridate)

scdb_case <- read_csv("data/scdb-case.csv")
```

```
Rows: 29227 Columns: 52
── Column specification
────────────────────────────────────────────────────────
Delimiter: ","
chr (14): caseId, docketId, caseIssuesId, dateDecision, usCite, sctCite,
led...
dbl (38): decisionType, term, naturalCourt, petitioner, petitionerState,
res...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

```r
scdb_oral <- scdb_case |>
  filter(decisionType == 1, !is.na(dateDecision)) |>
  mutate(
    dateDecision = mdy(dateDecision)
  )

scdb_oral <- scdb_oral |>
  mutate(
    month = month(dateDecision, label = TRUE, abbr = TRUE),
    month_num = month(dateDecision)
  )

decisions_by_month <- scdb_oral |>
  group_by(term, month_num, month) |>
  summarise(
    n_decisions = n(),
    .groups = "drop"
  )

nrow(decisions_by_month)
```

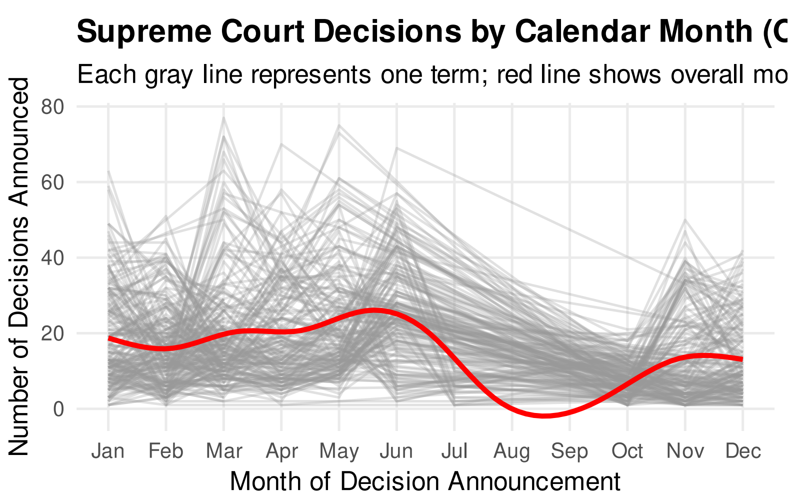
```
[1] 1478
```

```r
ggplot(decisions_by_month, aes(x = month_num, y = n_decisions, group = term))
+
```

```
  geom_line(alpha = 0.3, color = "gray60") +
  geom_smooth(aes(group = 1), color = "red", se = FALSE, size = 1.2) +
  scale_x_continuous(
    breaks = 1:12,
    labels = month.abb
  ) +
  labs(
    title = "Supreme Court Decisions by Calendar Month (Orally Argued Cases)",
    subtitle = "Each gray line represents one term; red line shows overall
monthly trend",
    x = "Month of Decision Announcement",
    y = "Number of Decisions Announced"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold"),
    panel.grid.minor = element_blank()
  )
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```



Exercise 5

```
library(tidyverse)

vote <- read_csv("data/scdb-vote.csv") |> rename_all(tolower)
```

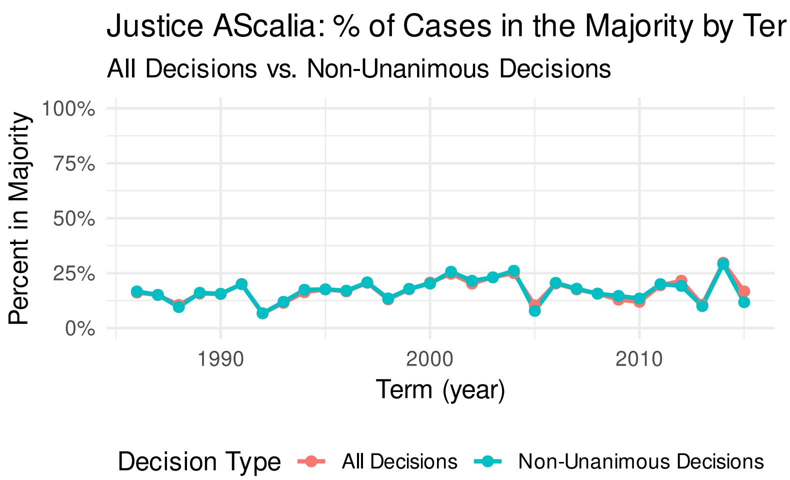
```
Rows: 255857 Columns: 13
── Column specification
────────────────────────────────────────────────────────
Delimiter: ","
chr (5): caseId, docketId, caseIssuesId, voteId, justiceName
dbl (8): term, justice, vote, opinion, direction, majority,
firstAgreement, ...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

```
case <- read_csv("data/scdb-case.csv") |> rename_all(tolower)
```

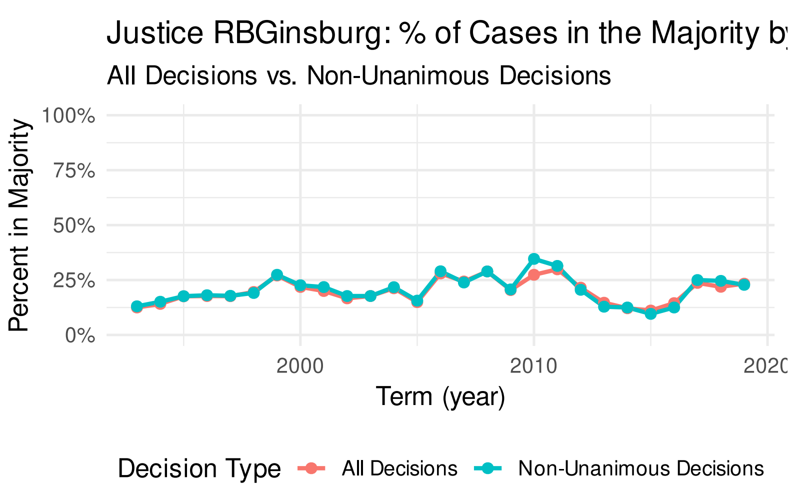
```
Rows: 29227 Columns: 52
── Column specification
────────────────────────────────────────────────────────
Delimiter: ","
chr (14): caseId, docketId, caseIssuesId, dateDecision, usCite, sctCite,
led...
dbl (38): decisionType, term, naturalCourt, petitioner, petitionerState,
res...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

```
vote <- vote |> select(caseid, justicename, majority, term)
case <- case |> select(caseid, term, datedecision)

maj_per_case <- vote |>
  mutate(majority_vote = majority == 1) |>
  group_by(caseid) |>
  summarise(majvotes = sum(majority_vote), .groups = "drop")

joined <- vote |>
  left_join(maj_per_case, by = "caseid") |>
  mutate(majority_vote = if_else(majority == 1, 1L, 0L))

plot_majority_rate <- function(justice_abbrev) {
  df <- joined |>
    filter(justicename == justice_abbrev, !is.na(term)) |>
    group_by(term) |>
    summarise(
      total_cases = n(),
```

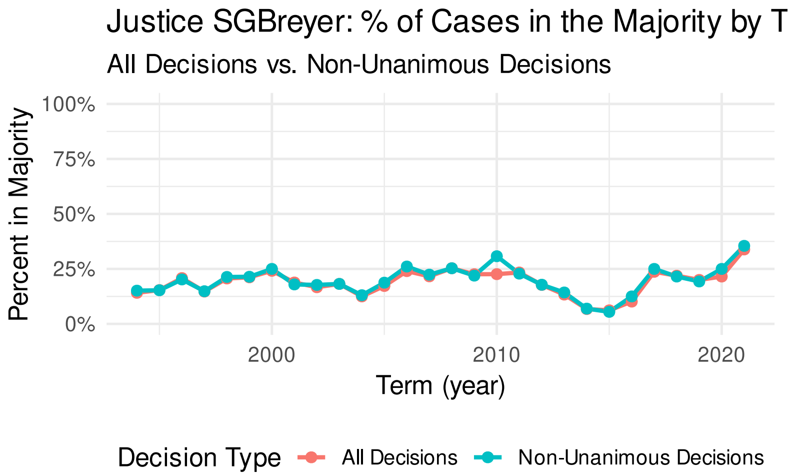
```
      majority_cases = sum(majority_vote, na.rm = TRUE),
      pct_majority_all = (majority_cases / total_cases) * 100,
      non_unanimous_cases = sum(majvotes < 9, na.rm = TRUE),
      majority_non_unanimous = sum(majority_vote == 1 & majvotes < 9, na.rm =
TRUE),
      pct_majority_nonunanimous = if_else(
        non_unanimous_cases > 0,
        (majority_non_unanimous / non_unanimous_cases) * 100,
        NA_real_
      ),
      .groups = "drop"
    ) |>
    pivot_longer(
      c(pct_majority_all, pct_majority_nonunanimous),
      names_to  = "category",
      values_to = "percent"
    ) %>%
    mutate(category = recode(
      category,
      pct_majority_all = "All Decisions",
      pct_majority_nonunanimous = "Non-Unanimous Decisions"
    ))

  ggplot(df, aes(x = term, y = percent, color = category)) +
    geom_line(linewidth = 1) +
    geom_point(size = 2) +
    scale_y_continuous(labels = scales::percent_format(scale = 1), limits =
c(0, 100)) +
    labs(
      title = paste0("Justice ", justice_abbrev, ": % of Cases in the Majority
by Term"),
      subtitle = "All Decisions vs. Non-Unanimous Decisions",
      x = "Term (year)",
      y = "Percent in Majority",
      color = "Decision Type"
    ) +
    theme_minimal(base_size = 13) +
    theme(legend.position = "bottom")
}

plot_majority_rate("AScalia")     # Antonin Scalia
```




```r
plot_majority_rate("RBGinsburg")  # Ruth Bader Ginsburg
```




```r
plot_majority_rate("SGBreyer")    # Stephen Breyer
```


Justice SGBreyer: % of Cases in the Majority by T
All Decisions vs. Non-Unanimous Decisions
Percent in Majority
100%
75%
50%
25%
0%
2000
2010
2020
Term (year)
Decision Type
All Decisions
Non-Unanimous Decisions